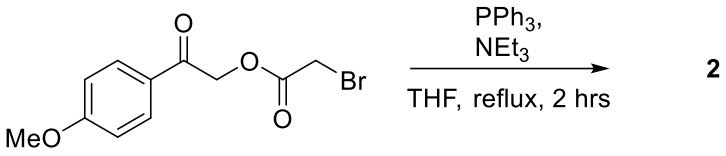
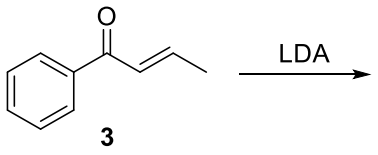
Question 1: Provide the reaction intermediates and products for the following reaction

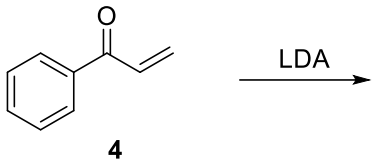


1

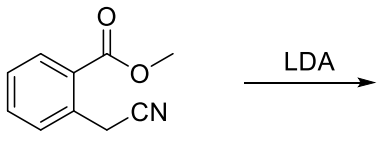
Question 2: Provide the corresponding enolate from the following compounds, if no enolate formed, place NA in the answer.



3



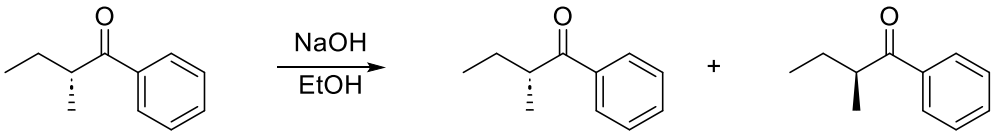
4



5

Question 3

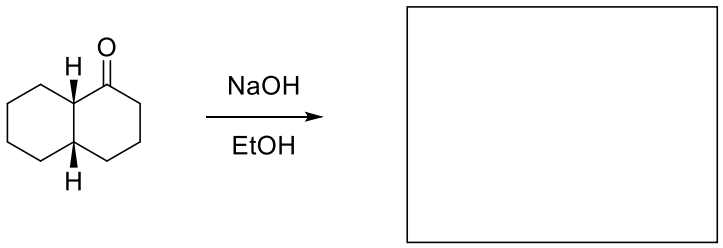
a. When a solution of (R)-(+)-2-methyl-1-phenylbutan-1-one in aqueous ethanol is treated with a base, the solution gradually loses its optical activity. Isolation of the ketones shows it has been racemized. Propose a mechanism for this transformation



6 (R) **6 (R)** **6 (S)**

50:50

b. Based on the finding in part a, predict the product for the following reaction

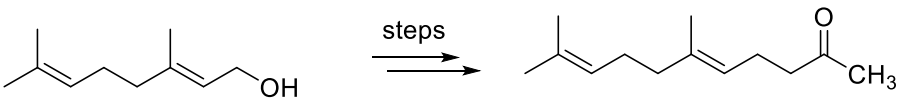


7
$C_{10}H_{16}O$

8
$C_{10}H_{16}O$

Question 4

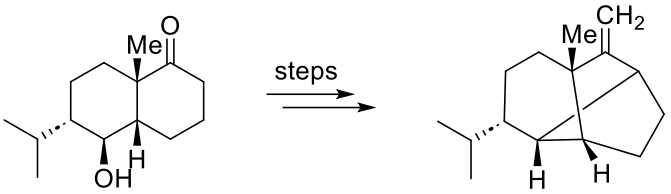
a. How could you convert geraniol to geranylacetone? More than 1 step may be required



9
geraniol

10
geranylacetone

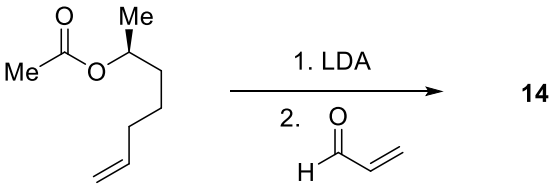
b. Propose a synthesis of a natural product Sativene from ketone **11**



11 **12**
sativene

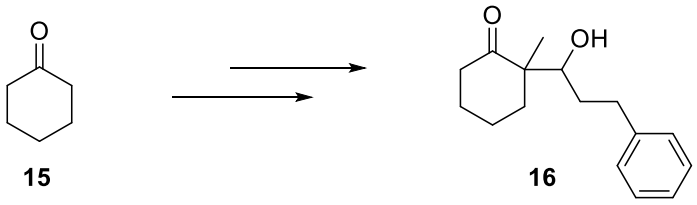
Question 5: Provide the reagents or products for the following transformations

a.



13 **14**

b.



15 **16**